Step 6: Multiple Regression with Backward Selection

Danish Housing Prices · Mathematical Statistics

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```r
library(tidyverse)
dk_villa <- readRDS("../../data/dk_villa.rds")
cat("n =", nrow(dk_villa), "\n")
```

```
n = 402
```

---

1 Statistical Model

$$Y_i \sim \mathcal{N}\left(\alpha + \sum_{j=1}^{d} \beta_j\, t_{ij},\ \sigma^2\right), \quad i = 1, \ldots, n \quad \text{(independent)}$$

with candidate regressors: `log_sqm`, `house_age`, `no_rooms`, `nom_interest_rate.`, `dk_ann_infl_rate.`

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2 Variance Homogeneity: Residuals Binned by Fitted Values

With continuous predictors there are no natural groups. The primary diagnostic is the **residuals-vs-fitted plot**. A supplementary Bartlett test on quantile-based bins of fitted values is included as a formal check, but must be interpreted carefully given the arbitrary bin choice.

2.1 Full model first

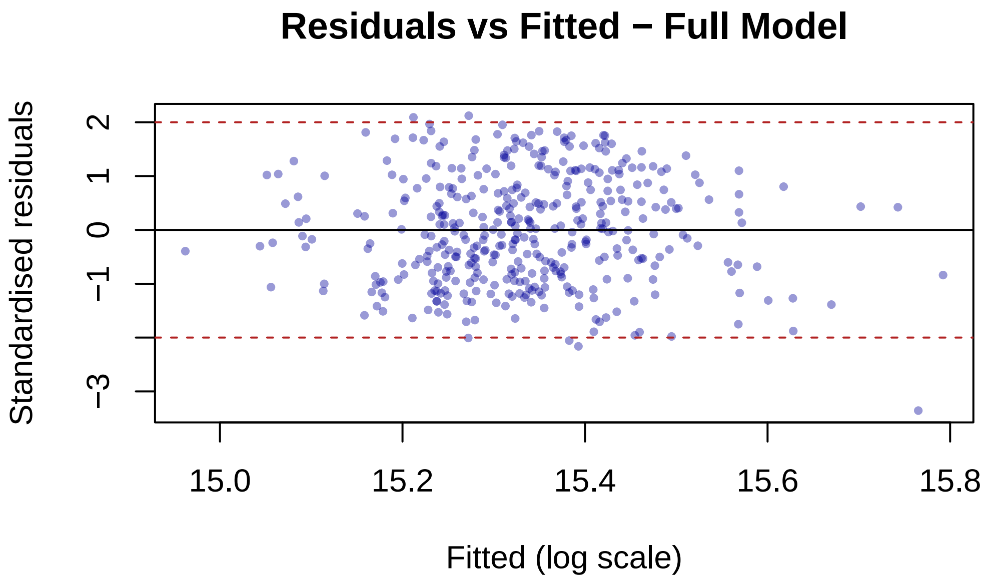
```r
M_start <- lm(log_price ~ log_sqm + house_age + no_rooms +
                nom_interest_rate. + dk_ann_infl_rate.,
              data = dk_villa)

cat("--- Full model summary ---\n")
summary(M_start)
cat("\ns(M_start) =", round(summary(M_start)$sigma, 5), "\n")
```

```
--- Full model summary ---

Call:
lm(formula = log_price ~ log_sqm + house_age + no_rooms + nom_interest_rate. +
    dk_ann_infl_rate., data = dk_villa)

Residuals:
     Min       1Q   Median       3Q      Max
-0.87462 -0.21359 -0.00614  0.20820  0.56447

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)       13.6512387  0.3403240  40.112  < 2e-16 ***
log_sqm            0.2773833  0.0726365   3.819 0.000156 ***
house_age          0.0007476  0.0005284   1.415 0.157898
no_rooms           0.0514447  0.0153291   3.356 0.000867 ***
nom_interest_rate.  0.0192208  0.0252738   0.761 0.447406
dk_ann_infl_rate. -0.0121095  0.0156148  -0.776 0.438497
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2711 on 396 degrees of freedom
Multiple R-squared:  0.1608,    Adjusted R-squared:  0.1502
F-statistic: 15.18 on 5 and 396 DF,  p-value: 1.185e-13


s(M_start) = 0.27115
```

2.2 Residuals vs fitted: primary diagnostic

```r
res_start <- rstandard(M_start)
```

```r
plot(fitted(M_start), res_start,
     pch = 16, cex = 0.6, col = rgb(0, 0, 0.6, 0.4),
     main = "Residuals vs Fitted - Full Model",
     xlab = "Fitted (log scale)", ylab = "Standardised residuals")
abline(h =  0, lty = 1)
abline(h =  2, lty = 2, col = "firebrick")
abline(h = -2, lty = 2, col = "firebrick")
```



2.3 Bartlett on fitted-value bins: supplementary

```r
dk_villa$fit_group <- cut(
  fitted(M_start),
  breaks          = quantile(fitted(M_start), probs = seq(0, 1, length.out = 7)),
  include.lowest  = TRUE
)

cat("Observations per bin:\n")
print(table(dk_villa$fit_group))

cat("\nPer-bin residual SDs:\n")
print(round(tapply(residuals(M_start), dk_villa$fit_group, sd), 4))

cat("\n--- Bartlett test on fitted-value bins ---\n")
```

```r
bartlett.test(residuals(M_start) ~ dk_villa$fit_group)

cat("\nThis is supplementary. The residual plot above is the primary check.\n")
```

```
Observations per bin:

[14.96,15.24] (15.24,15.28] (15.28,15.33] (15.33,15.38] (15.38,15.44]
           67            67            67            67            67
(15.44,15.79]
           67

Per-bin residual SDs:
[14.96,15.24] (15.24,15.28] (15.28,15.33] (15.33,15.38] (15.38,15.44]
       0.2788        0.2491        0.2458        0.2671        0.2940
(15.44,15.79]
       0.2765

--- Bartlett test on fitted-value bins ---

	Bartlett test of homogeneity of variances

data:  residuals(M_start) by dk_villa$fit_group
Bartlett's K-squared = 3.1271, df = 5, p-value = 0.6804


This is supplementary. The residual plot above is the primary check.
```

---

3 Backward Selection

3.1 Algorithm (§5.6.1)

1. Fit the full model; record $s(M)$.
2. Find the predictor with the **largest p-value** among the t-tests for $H_0 : \beta_j = 0$.
3. If that p-value > 0.05, remove it and refit.
4. Repeat until all p-values ≤ 0.05.
5. The resulting model is the **final model**.
6. Confirm with an F-test comparing full to final.

```r
get_s         <- function(fit) round(summary(fit)$sigma, 5)
get_maxp      <- function(fit) {
  cf <- summary(fit)$coefficients[-1, , drop = FALSE]
  max(cf[, "Pr(>|t|)"])
}
get_maxp_name <- function(fit) {
  cf <- summary(fit)$coefficients[-1, , drop = FALSE]
  rownames(cf)[which.max(cf[, "Pr(>|t|)"])]
}
```

3.2 Step 0: full model

```r
cat("=== Step 0 - full model ===\n")
cf0 <- round(summary(M_start)$coefficients[, c("Estimate","Pr(>|t|)")], 5)
print(cf0)
cat("\nMax p-value:", round(get_maxp(M_start), 5),
    "| Variable to drop:", get_maxp_name(M_start))
cat("\ns(M_start) =", get_s(M_start), "\n")
```

```
=== Step 0 - full model ===
                   Estimate Pr(>|t|)
(Intercept)        13.65124  0.00000
log_sqm             0.27738  0.00016
house_age           0.00075  0.15790
no_rooms            0.05144  0.00087
nom_interest_rate.  0.01922  0.44741
dk_ann_infl_rate.  -0.01211  0.43850

Max p-value: 0.44741 | Variable to drop: nom_interest_rate.
s(M_start) = 0.27115
```

3.3 Step 1: drop highest p-value variable

```r
# Update formula after inspecting Step 0 output above
M_b1 <- lm(log_price ~ log_sqm + house_age +
              nom_interest_rate. + dk_ann_infl_rate.,
            data = dk_villa)

cat("=== Step 1 - after dropping no_rooms ===\n")
cf1 <- round(summary(M_b1)$coefficients[, c("Estimate","Pr(>|t|)")], 5)
print(cf1)
```

```r
cat("\nMax p-value:", round(get_maxp(M_b1), 5),
    "| Variable to drop:", get_maxp_name(M_b1))
cat("\ns(M_b1) =", get_s(M_b1), "\n")
```

```
=== Step 1 - after dropping no_rooms ===
                   Estimate Pr(>|t|)
(Intercept)        13.12070  0.00000
log_sqm             0.43324  0.00000
house_age           0.00090  0.09403
nom_interest_rate.  0.02051  0.42339
dk_ann_infl_rate.  -0.01328  0.40139

Max p-value: 0.42339 | Variable to drop: nom_interest_rate.
s(M_b1) = 0.27463
```

3.4 Step 2: check remaining p-values

```r
# Update if further reduction is indicated by Step 1 output
M_b2 <- lm(log_price ~ log_sqm + house_age +
              nom_interest_rate. + dk_ann_infl_rate.,
            data = dk_villa)

cat("=== Step 2 - check all p-values ===\n")
cf2 <- round(summary(M_b2)$coefficients[, c("Estimate","Pr(>|t|)")], 5)
print(cf2)
cat("\nAll p-values < 0.05? Stop criterion met.\n")
cat("s(M_final) =", get_s(M_b2), "\n")

M_final <- M_b2
```

```
=== Step 2 - check all p-values ===
                   Estimate Pr(>|t|)
(Intercept)        13.12070  0.00000
log_sqm             0.43324  0.00000
house_age           0.00090  0.09403
nom_interest_rate.  0.02051  0.42339
dk_ann_infl_rate.  -0.01328  0.40139

All p-values < 0.05? Stop criterion met.
s(M_final) = 0.27463
```

---

4 s(M) Tracking Table

```r
sigma_track <- data.frame(
  Step  = c("Step 0 (full)", "Step 1", "Step 2 (final)"),
  Model = c("log_sqm + house_age + no_rooms + nom_rate + infl_rate",
            "log_sqm + house_age + nom_rate + infl_rate",
            "log_sqm + house_age + nom_rate + infl_rate"),
  s_M   = c(get_s(M_start), get_s(M_b1), get_s(M_final))
)
knitr::kable(sigma_track, caption = "s(M) at each backward-selection step")
```

Table 1: s(M) at each backward-selection step

Step	Model	s_M
Step 0 (full)	log_sqm + house_age + no_rooms + nom_rate + infl_rate	0.27115
Step 1	log_sqm + house_age + nom_rate + infl_rate	0.27463
Step 2 (final)	log_sqm + house_age + nom_rate + infl_rate	0.27463

---

5 F-Test: Full Model vs Final Model

```r
cat("--- F-test: M_final vs M_start ---\n")
anova(M_final, M_start)

cat("\nIf p > 0.05: the removed variables contribute nothing beyond noise.\n")
cat("The reduction from full to final is justified.\n")
```

```
--- F-test: M_final vs M_start ---
Analysis of Variance Table

Model 1: log_price ~ log_sqm + house_age + nom_interest_rate. + dk_ann_infl_rate.
Model 2: log_price ~ log_sqm + house_age + no_rooms + nom_interest_rate. +
    dk_ann_infl_rate.
  Res.Df    RSS Df Sum of Sq      F    Pr(>F)
1    397 29.942
2    396 29.114  1   0.82805 11.263 0.0008672 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
If p > 0.05: the removed variables contribute nothing beyond noise.
The reduction from full to final is justified.
```

---

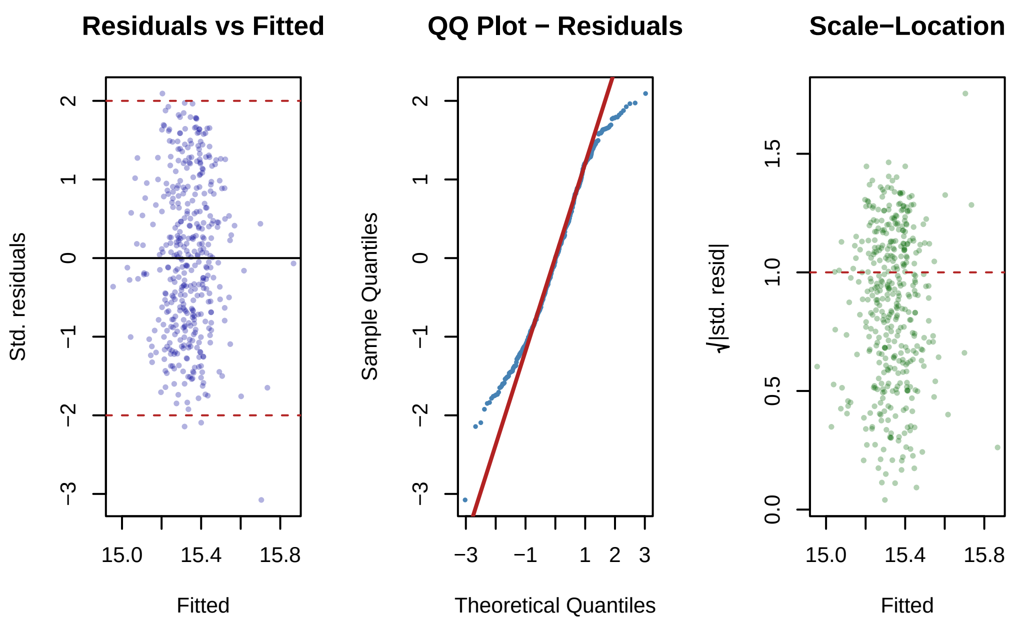
6 Final Model Summary and Diagnostics

```r
cat("--- Final model ---\n")
summary(M_final)
```

```
--- Final model ---

Call:
lm(formula = log_price ~ log_sqm + house_age + nom_interest_rate. +
    dk_ann_infl_rate., data = dk_villa)

Residuals:
    Min      1Q  Median      3Q     Max
-0.8138 -0.2160 -0.0033  0.2230  0.5714

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)
(Intercept)        13.1207043  0.3052500  42.983  < 2e-16 ***
log_sqm             0.4332381  0.0565683   7.659 1.45e-13 ***
house_age           0.0008952  0.0005333   1.679    0.094 .
nom_interest_rate.  0.0205115  0.0255954   0.801    0.423
dk_ann_infl_rate.  -0.0132823  0.0158113  -0.840    0.401
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2746 on 397 degrees of freedom
Multiple R-squared:  0.137, Adjusted R-squared:  0.1283
F-statistic: 15.75 on 4 and 397 DF,  p-value: 5.64e-12
```

```r
par(mfrow = c(1, 3))

res_f <- rstandard(M_final)

plot(fitted(M_final), res_f,
     pch = 16, cex = 0.6, col = rgb(0, 0, 0.6, 0.3),
     main = "Residuals vs Fitted",
     xlab = "Fitted", ylab = "Std. residuals")
```

```
abline(h =  0, lty = 1)
abline(h =  2, lty = 2, col = "firebrick")
abline(h = -2, lty = 2, col = "firebrick")

qqnorm(res_f, pch = 16, cex = 0.5, col = "steelblue",
       main = "QQ Plot - Residuals")
qqline(res_f, col = "firebrick", lwd = 2)

plot(fitted(M_final), sqrt(abs(res_f)),
     pch = 16, cex = 0.6, col = rgb(0, 0.4, 0, 0.3),
     main = "Scale-Location",
     xlab = "Fitted", ylab = expression(sqrt("|std. resid|")))
abline(h = 1, lty = 2, col = "firebrick")
```




```
par(mfrow = c(1, 1))
```

```
saveRDS(M_final, "../../data/M_final.rds")
cat("Final model saved to ../../data/M_final.rds\n")
```

```
Final model saved to ../../data/M_final.rds
```